Inputs
Active Client Layer Updates
candidate updates from active clients
$\Delta W_k^{(l)}$
RawScore from Stage 3
trust-weighted permission score
$RawScore_k$
Control & Decision
Privacy Sensitivity
exposure risk of layer 1
$S_{\text{priv}}^{(l)}$
Utility Sensitivity
contribution to global direction
$S_{\text{util}}^{(l)}$
Security Deviation Sensitivity
deviation from trusted reference
$S_{\text{sec}}^{(l)}$
risk fusion
Total Defense Requirement
weighted fusion of three signals
$S_{\text{total}}^{(l)} = w_p S_{\text{priv}}^{(l)} + w_u S_{\text{util}}^{(l)} + w_s S_{\text{sec}}^{(l)}$
threshold control
radius control
Layer Threshold
admission barrier for this layer
$\theta^{(l)}$
Clip Radius
maximum safe update magnitude
$C^{(l)}$
Constrained Execution
survivor filtering
Survivor Selection
keep only qualified clients
$\Phi^{(l)} = \{k : RawScore_k \geq \theta^{(l)}\}$
reweight survivors
Weight Re-normalization
redistribute weights over survivors
$\sum \tilde{w}_k^{(l)} = 1$
clip magnitude
Dynamic L2 Clipping
project updates into safe radius
$\|\hat{\Delta W}_k^{(l)}\|_2 \leq C^{(l)}$
aggregate securely
Secure Layer Aggregation
aggregate filtered clipped survivors
$\Delta W_{\text{global}}^{(l)} = \sum \tilde{w}_k^{(l)} \hat{\Delta W}_k^{(l)}$
Global Layer Update
applied to global model
$\Delta W_{\text{global}}^{(l)}$